

INFO-H512 Project

1. Project Overview

The goal of the project is to **study in depth one state-of-the-art AI method** presented in the seminar. Students will develop a strong understanding of a selected topic through literature review and practical implementation.

2. Project Description

Problem Formulation

Each group must select a **topic** presented by one of the seminar speakers (e.g., *Counterfactual Explanations*) and identify a **specific method** within this topic to explore in detail (e.g., *Feasible Actions for Counterfactual Explanations (FACE)*).

The project must define a clear **research question** or problem statement that requires a software implementation to address.

Example directions include:

- Reproducing results from a paper;
- Applying the method to a new dataset or problem setting;
- Extending or modifying the method in a meaningful way.

When feasible, groups are encouraged to interact with the seminar speaker for guidance or feedback.

Literature Review

The project must include a **targeted literature review** of the selected topic. A list of references for most topics is available on UV as a starting point.

This should include:

- A concise **taxonomy** of the research area;
 - Identification of **seminal and recent works**;
 - A **critical discussion** of assumptions, strengths, limitations, and open challenges;
 - Clear **positioning** of the selected method within the broader literature.
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Evaluation and Analysis

The project must include a software implementation together with **experimental evaluation** and **critical analysis** addressing the research question.

Experimental evaluation may include:

- Performance metrics appropriate to the task;
- Comparison with baselines or simpler methods;
- Analysis across datasets or settings.

Critical analysis should discuss:

- Strengths and limitations of the method;
- Key assumptions and their implications;
- Potential improvements or open research questions.

Results should be presented clearly using **tables, figures, or plots** where appropriate.

Note: Python is preferred, but alternative tools or languages may be used with justification. Use of high-level libraries requires a clear understanding of the underlying methods.

3. Evaluation Rubric

The project will be graded out of **10 points**, according to the following rubric.

Aspect	Excellent (A)	Good (B)	Satisfactory (C)
Problem Formulation (2 pts)	Clear, well-motivated research question; precise problem definition; strong connection to seminar content and chosen method.	Problem is relevant but lacks full clarity or precise formulation; connection to method is present but not fully justified.	Problem is vague or weakly specified; unclear connection to seminar content or method.
Literature Review (3 pts)	Comprehensive and well-structured review; clear taxonomy; strong critical analysis; excellent positioning of the method.	Adequate review with some structure; limited depth in analysis or positioning.	Minimal or superficial review; lacks structure and critical insight.
Evaluation and	Well-designed experiments with appropriate metrics; results clearly presented;	Experiments and analysis are present	Minimal evaluation; weak or mostly

Analysis (4 pts)	insightful and thorough analysis of assumptions, trade-offs, and limitations.	but somewhat limited in scope, rigor, or depth.	descriptive analysis; limited insight.
Report Quality (1 pt)	Clear, concise, and well-organized report; effective use of figures/tables.	Generally clear but could be better organized or more concise.	Difficult to follow; lacks clarity or structure.

4. Submission Instructions

Submit your project via **UV** by **June 5** as a single PDF report of **up to 4 pages** following the [AAAI 2026 formatting instructions](#).

Notes:

- References may appear on additional pages.
- The report must include a link to a **GitHub repository** containing the project code and clear instructions for reproducing the results.
- During the week of **June 8**, each group will give a **short presentation** of their work as part of the final oral exam.